Norges miljø- og biovitenskapelige universitet

TEL200 – Introduction to Robotics

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Chapter 3: Time and Motion

Some basic notions

- Pose: the position and orientation ξ of an object (Ch 2.)
  - Can be used for defining start and end poses, A and B.
- Path: a varying pose $\xi(s)$, for some parameter $s \in [s_0, S_T]$
  - Takes us from A to B but has no notion of time
- Trajectory: a path with specified timing, $\xi(s(t))$, for $t \in [0, T]$
  - Defines the time-evolution, or speed along the path from A to B
- Note: The notions of path and trajectory can also be generalized to motion in the configuration space or joint space of a robot.

Time-Varying Pose

- ***How*** to describe the rate of change of ***pose*** which has both a translational and rotational velocity component ?

- The translational velocity is straightforward: it is the rate of change of the ***position*** of the origin of the coordinate frame.

- ***Rotational velocity*** is a little more complex.

Derivative of Pose

- There are ***many ways*** to represent the orientation matrix of a coordinate frame but most convenient for present purposes is the ***exponential form*** *(see Sect. 2.3.1.6)*:

$$ {}^{A}\boldsymbol{R}_{B}(t) = \mathrm{e}^{\left[{}^{A}\hat{\omega}(t)\right]_{\times}\theta(t)} \in \mathrm{SO}(3) $$

where:

${}^{A}\hat{\omega}(t)$ denotes **rotation axis** with respect to frame $\{A\}$

$\theta(t)$ is the ***rotational angle***,

$[\cdot]_{\times}$ is a ***skew-symmetric*** matrix.

Derivative of Pose

- At an instant in time t we will assume that the ***axis*** has a fixed direction, and the ***frame*** is rotating around the axis.
- The ***derivative*** with respect to time is

$$
\begin{aligned}
{}^{A}\dot{\mathbf{R}}_{B}(t) &= \dot{\theta}(t)\left[{}^{A}\hat{\boldsymbol{\omega}}_{B}\right]_{\times} \mathrm{e}^{\theta(t)\left[{}^{A}\hat{\boldsymbol{\omega}}_{B}\right]_{\times}} \in \mathbb{R}^{3\times 3} \\
&= \dot{\theta}(t)\left[{}^{A}\hat{\boldsymbol{\omega}}_{B}\right]_{\times} {}^{A}\mathbf{R}_{B}(t)
\end{aligned}
$$

which we write succinctly as

$$
{}^{A}\dot{\mathbf{R}}_{B} = \left[{}^{A}\boldsymbol{\omega}_{B}\right]_{\times} {}^{A}\mathbf{R}_{B} \in \mathbb{R}^{3\times 3} \tag{3.1}
$$

where ${}^{A}\omega_{B} = \dot{\theta}\,{}^{A}\hat{\boldsymbol{\omega}}_{B}$ is the angular velocity of frame $\{\mathrm{B}\}$ w.r.t. frame $\{A\}$.

Derivative of Pose

- Angular velocity:

$$^{A}\boldsymbol{\omega} = {}^{A}\boldsymbol{R}_{B}\,{}^{B}\boldsymbol{\omega}$$

$^{A}\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)$ rate of rotation in frame {A}

$\|{}^{A}\boldsymbol{\omega}\|$ magnitude rate of rotation

$^{B}\boldsymbol{\omega}$ rate of rotation in frame {B}

Derivative of Pose

- The ***derivative of pose*** can be determined by expressing pose as a homogeneous transformation matrix

$$ {}^{A}\mathbf{T}_{B} = \begin{pmatrix} {}^{A}\mathbf{R}_{B} & {}^{A}\boldsymbol{t}_{B} \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix} \in \mathbf{SE}(3) $$

and taking the derivative with respect to time and substituting Eq. (3.1) gives

$$ {}^{A}\dot{\mathbf{T}}_{B} = \begin{pmatrix} {}^{A}\dot{\mathbf{R}}_{B} & {}^{A}\dot{\boldsymbol{t}}_{B} \\ \mathbf{0}_{1\times 3} & 0 \end{pmatrix} = \begin{pmatrix} [{}^{A}\boldsymbol{\omega}_{B}]_{\times}\,{}^{A}\mathbf{R}_{B} & {}^{A}\dot{\boldsymbol{t}}_{B} \\ \mathbf{0}_{1\times 3} & 0 \end{pmatrix} \in \mathbb{R}^{4\times 4} \tag{3.4} $$

Derivative of Pose

- The ***rate of change*** can be described in terms of the current orientation ${}^{A}R_{B}$ and *two* velocities:
  - The linear or translational velocity ${}^{A}\boldsymbol{v}_{B} = {}^{A}\dot{\boldsymbol{t}}_{B}$ is the velocity of $\{B\}$ w.r.t. $\{A\}$;
  - The angular velocity ${}^{A}\boldsymbol{\omega}_{B}$ we have already introduced ;
- We can ***combine*** these two velocity vectors to create the spatial velocity vector

$$ {}^{A}\nu_{B} = \left({}^{A}\boldsymbol{v}_{B}, {}^{A}\boldsymbol{\omega}_{B} \right) \in \mathbb{R}^{6} $$

which is the ***instantaneous velocity*** of frame $\{B\}$ with respect to $\{A\}$.

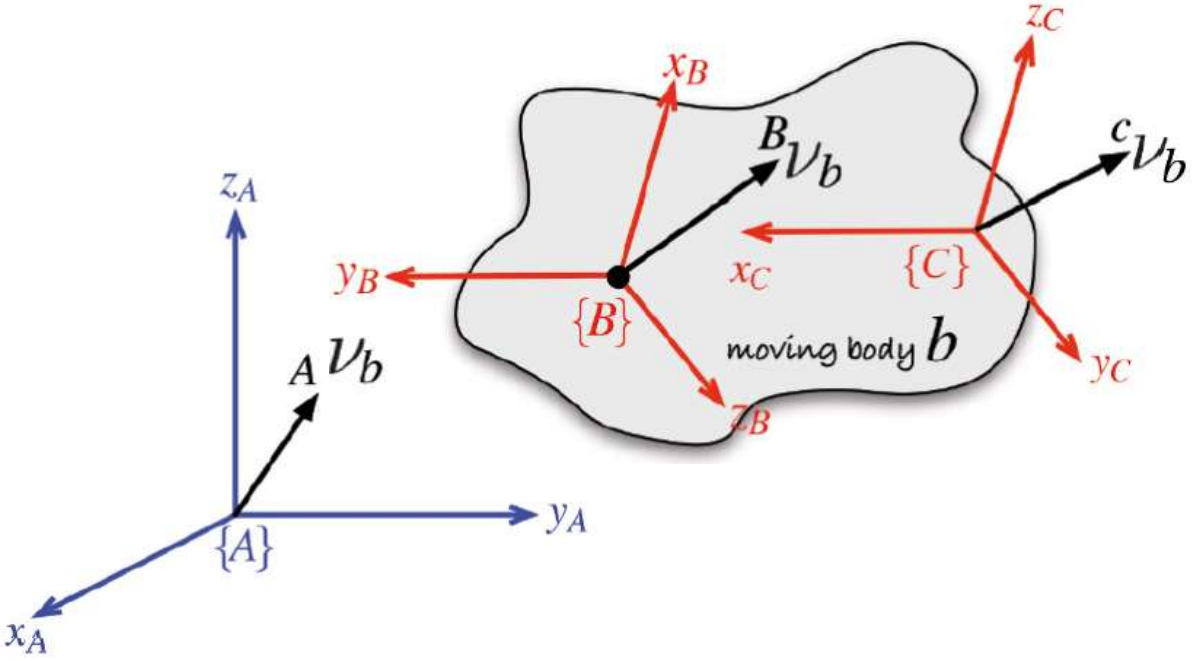
Transforming Spatial Velocities

- The ***velocity*** of a moving body can be expressed with respect to a world reference frame $\{A\}$ or the moving body frame $\{B\}$
- Spatial velocities are linearly related by:

$$ {}^{A}\nu = \begin{pmatrix} {}^{A}\boldsymbol{R}_B & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & {}^{A}\boldsymbol{R}_B \end{pmatrix} {}^{B}\nu = {}^{A}\boldsymbol{J}_B\left({}^{A}\xi_B\right){}^{B}\nu \qquad (3.5) $$

where $\boxed{{}^{A}\xi_B \sim ({}^{A}\boldsymbol{R}_B, {}^{A}\boldsymbol{t}_B)}$ and $\boxed{{}^{A}\boldsymbol{J}_B(\cdot)}$ is a Jacobian or interaction matrix.

Representing the spatial velocity of a moving body b with respect to various coordinate frames. Note that ν is a 6-dimensional vector



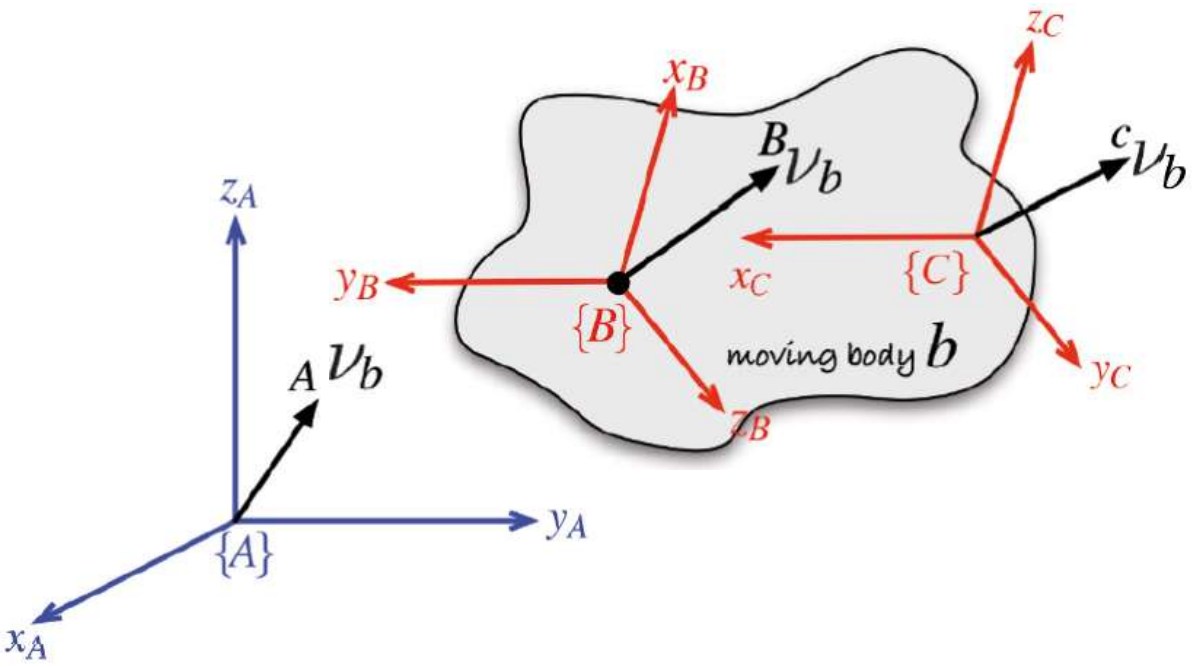
Transforming Spatial Velocities

- For the case where frame $\{C\}$ is also on the moving body the transformation becomes

$$ {}^{C}\nu = \begin{pmatrix} {}^{C}\boldsymbol{R}_B & \left[{}^{C}\boldsymbol{t}_B\right]_\times {}^{C}\boldsymbol{R}_B \\ \boldsymbol{0}_{3\times 3} & {}^{C}\boldsymbol{R}_B \end{pmatrix} {}^{B}\nu = \mathrm{Ad}\left({}^{C}\xi_B\right) {}^{B}\nu $$

and involves the adjoint matrix of the relative pose.

Representing the spatial velocity of a moving body b with respect to various coordinate frames. Note that ν is a 6-dimensional vector



Dynamics of Moving Bodies

- For **translational** motion Newton's second law describes, in the inertial frame, the acceleration of a particle with position x and mass m

$$m\,{}^{0}\ddot{\boldsymbol{x}}_B = {}^{0}\boldsymbol{f}_B \tag{3.13}$$

due to the applied force ${}^{0}f_{\mathrm{B}}$.

Dynamics of Moving Bodies

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- **Rotational** motion in **SO**(3) is described by Euler's equations of motion which relates the angular acceleration of the body in the body frame

$${}^{B}\mathbf{J}_B\,{}^{B}\dot{\boldsymbol{\omega}}_B + {}^{B}\boldsymbol{\omega}_B \times \left({}^{B}\mathbf{J}_B\,{}^{B}\boldsymbol{\omega}_B\right) = {}^{B}\boldsymbol{\tau}_B \tag{3.14}$$

$$\boldsymbol{J} = \begin{pmatrix} J_{xx} & J_{xy} & J_{xz} \\ J_{xy} & J_{yy} & J_{yz} \\ J_{xz} & J_{yz} & J_{zz} \end{pmatrix}$$

to the applied torque or moment ${}^{\mathrm{B}}\tau_{\mathrm{B}}$ and a positive-definite rotational inertia matrix ${}^{B}\mathbf{J}_B \in \mathbb{R}^{3\times 3}$

Transforming Forces and Torques

- The ***spatial velocity*** is a vector quantity that represents translational and rotational velocity
- Similarly, we can combine translational force and rotational torque (or moment) into a 6-vector that is called a ***wrench***

$$\boldsymbol{W} = (f_x, f_y, f_z, m_x, m_y, m_z) \in \mathbb{R}^6.$$

- Two wrenchs, ${}^{\mathrm{A}}\boldsymbol{w}$ and ${}^{\mathrm{B}}\boldsymbol{w}$ are defined with respect to the coordinate frame $\{\mathrm{A}\}$ and $\{\mathrm{B}\}$ respectively and both are applied at the origin of the frames.
- They are *equivalent* if they cause the same motion on the body

Transforming Forces and Torques

- The wrenches, ${}^{A}\boldsymbol{w}$ and ${}^{B}\boldsymbol{w}$ are related by

$$
{}^{A}\boldsymbol{w} = \begin{pmatrix} {}^{B}\mathbf{R}_{A} & \left[{}^{B}\boldsymbol{t}_{A}\right]_{\times}{}^{B}\mathbf{R}_{A} \\ \mathbf{0}_{3\times 3} & {}^{B}\mathbf{R}_{A} \end{pmatrix}^{\top} {}^{B}\boldsymbol{w} = \mathrm{Ad}^{\top}\left({}^{B}\mathbf{T}_{A}\right) {}^{B}\boldsymbol{w} \tag{3.15}
$$

which is ***similar*** to the spatial velocity transform of Eq. (3.5) but uses the transpose of the adjoint of the *inverse* relative pose.

Inertial Reference Frame

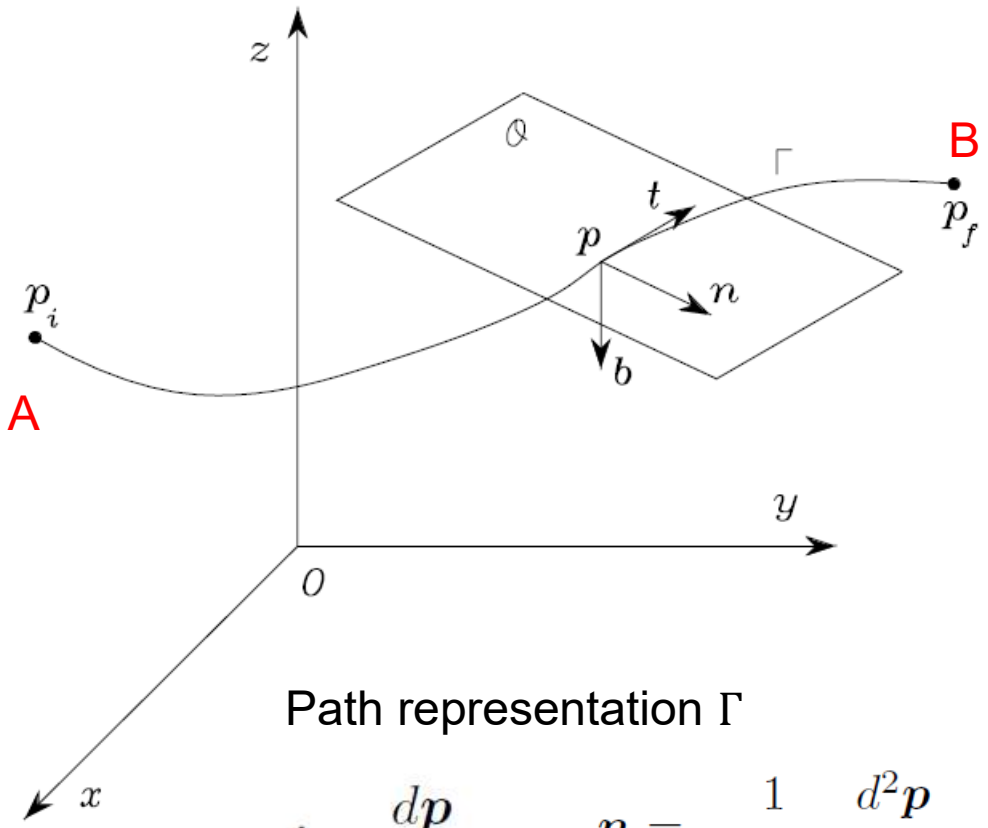
- Inertial Reference Frame is a reference frame that is not accelerating or rotating.
- If the reference frame $\{B\}$ is ***rotating*** with angular velocity ω about its origin, then Newton's second law Eq. (3.13) becomes:

$$m\left({}^{B}\dot{\boldsymbol{v}} + \underbrace{\boldsymbol{\omega} \times (\boldsymbol{\omega} \times {}^{B}\boldsymbol{p})}_{\text{centripetal}} + \underbrace{2\boldsymbol{\omega} \times {}^{B}\boldsymbol{v}}_{\text{Coriolis}} + \underbrace{\dot{\boldsymbol{\omega}} \times {}^{B}\boldsymbol{p}}_{\text{Euler}}\right) = {}^{0}\boldsymbol{f}$$

- ***Centripetal acceleration*** always acts inward toward the origin;
- If the point is moving, then ***Coriolis acceleration*** will be normal to its velocity;
- If rotational velocity is time varying, then ***Euler acceleration*** will be normal to the position vector.
- In robotics, «world coordinate frame» typically constitutes an Inertial Reference Frame

Creating Time-varying Pose

- In ***robotics*** we often need to generate a time-varying pose (i.e., a path) that moves ***smoothly*** in translation and rotation.



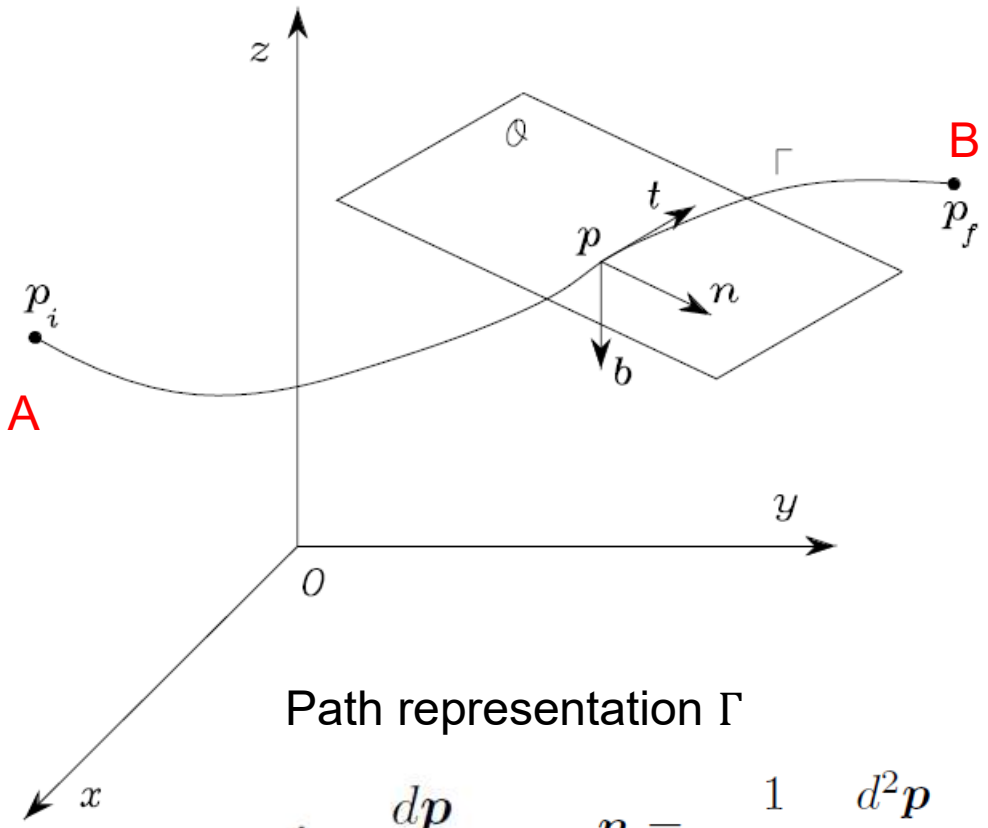
Path representation Γ

$$\boldsymbol{t} = \frac{d\boldsymbol{p}}{ds} \qquad \boldsymbol{n} = \frac{1}{\left\| \frac{d^2\boldsymbol{p}}{ds^2} \right\|} \frac{d^2\boldsymbol{p}}{ds^2}$$

$$\boldsymbol{b} = \boldsymbol{t} \times \boldsymbol{n}.$$

Creating Time-varying Pose

- In ***robotics*** we often need to generate a time-varying pose (i.e., a path) that moves ***smoothly*** in translation and rotation.
- A ***path*** is a spatial construct – a ***locus*** in space that leads from an initial pose A to a final pose B.



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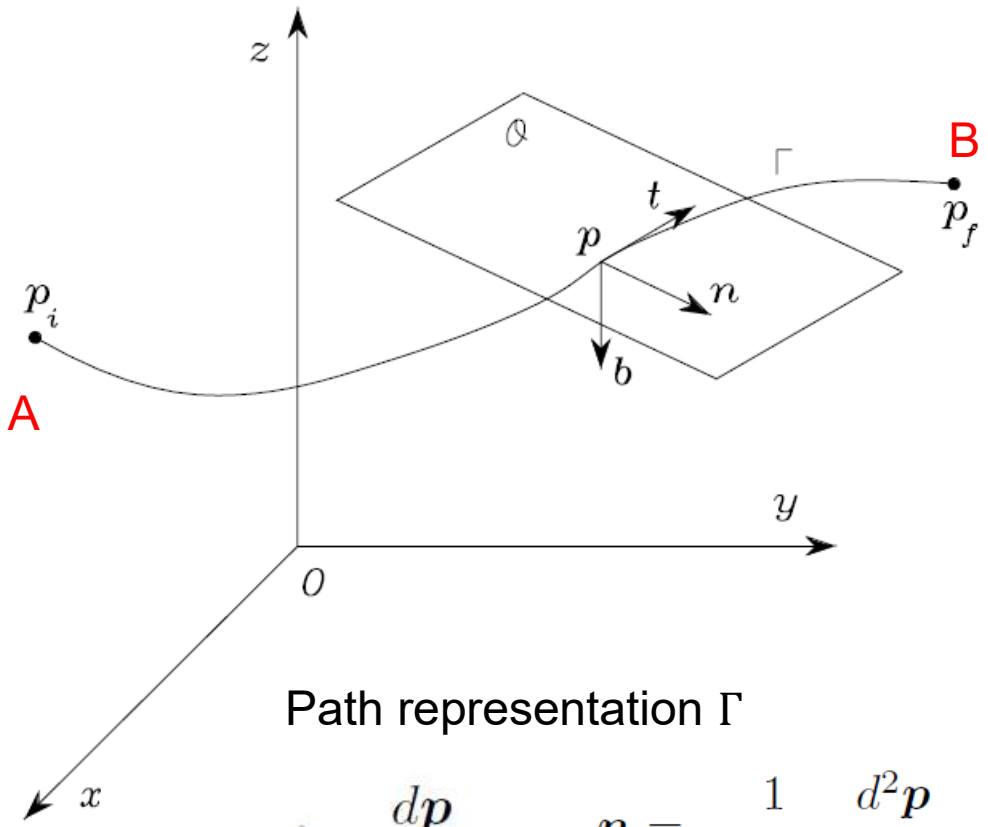
Creating Time-varying Pose

- In ***robotics*** we often need to generate a time-varying pose (i.e., a path) that moves ***smoothly*** in translation and rotation.
- A ***path*** is a spatial construct – a ***locus*** in space that leads from an initial pose A to a final pose B.
- A ***trajectory*** is a path with specified timing. For example, there is a ***path*** from A to B, but there is a ***trajectory*** from A to B in time $\Delta t = 10\ s$ or at velocity $\mathrm{v} = 2ms^{-1}$.



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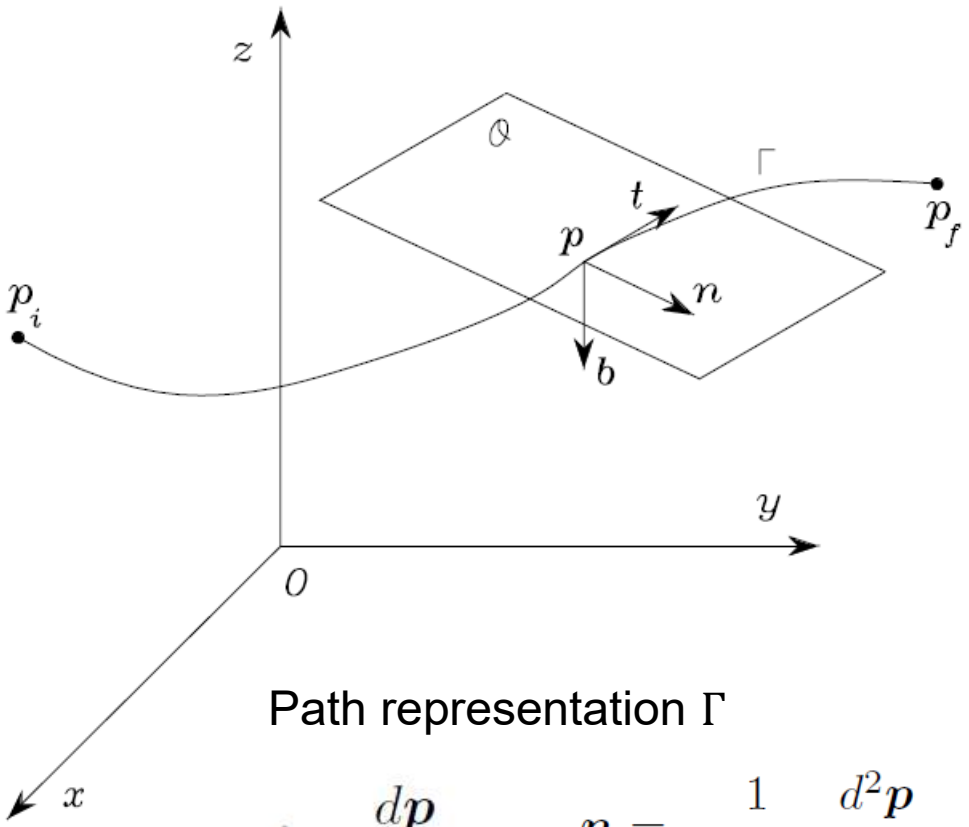
$$\boldsymbol{b} = \boldsymbol{t} \times \boldsymbol{n}.$$

Creating Time-varying Pose

- An important characteristic of a trajectory is that it is ***smooth*** – position and orientation vary smoothly with time t.
- We start by discussing how to generate ***smooth trajectories*** in one dimension.
- Then, we extend ***trajectory generation*** to the multi-dimensional case and next to piecewise-linear trajectories that visit a number of intermediate points without stopping.



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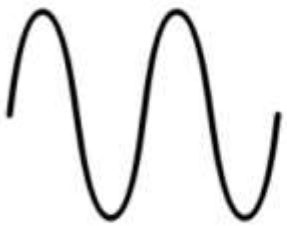
$$\boldsymbol{b} = \boldsymbol{t} \times \boldsymbol{n}.$$

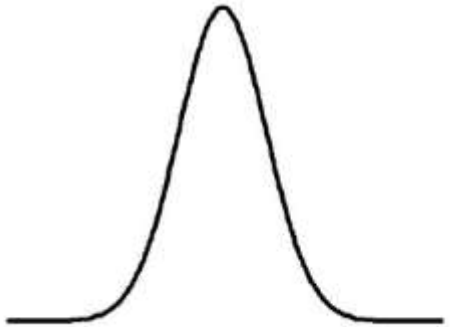
Smooth One-Dimensional Trajectories

- ***Smoothness*** in this context means that its first few temporal derivatives are ***continuous***.

Position	$x(t)$	are continuous
Velocity	$\dot{x}(t)$	
Acceleration	$\ddot{x}(t)$	
Jerk	$\dddot{x}(t)$	
and so on...		

- Examples: Sinusoid Gaussian Polynomial





Smooth One-Dimensional Trajectories

- Consider a ***scalar function*** of time $q(t)$, where time $t \in [0, T]$.
- Important characteristics of this function are that its ***initial*** and ***final*** value are specified and that it is smooth.

$$q(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

Smooth One-Dimensional Trajectories

- Consider a ***scalar function*** of time $q(t)$, where time $t \in [0, T]$.
- Important characteristics of this function are that its ***initial*** and ***final*** value are specified and that it is smooth.
- ***Smoothness*** in this context means that its first few temporal derivatives (i.e., $\dot{q}$ and $\ddot{q}$) are ***continuous***.

$$q(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

$$\dot{q}(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2Dt + E$$

$$\ddot{q}(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

Smooth One-Dimensional Trajectories

- Consider a ***scalar function*** of time $q(t)$, where time $t \in [0, T]$.
- Important characteristics of this function are that its ***initial*** and ***final*** value are specified and that it is smooth.
- ***Smoothness*** in this context means that its first few temporal derivatives (i.e., $\dot{q}$ and $\ddot{q}$) are ***continuous***.
- Typically, ***velocity*** and ***acceleration*** are required to be continuous and sometimes also the derivative of acceleration or ***jerk***.

$$q(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

$$\dot{q}(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2Dt + E$$

$$\ddot{q}(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

$$\begin{pmatrix} q_0 \\ q_T \\ \dot{q}_0 \\ \dot{q}_T \\ \ddot{q}_0 \\ \ddot{q}_T \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ T^5 & T^4 & T^3 & T^2 & T & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 5T^4 & 4T^3 & 3T^2 & 2T & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 20T^3 & 12T^2 & 6T & 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \end{pmatrix}$$

Smooth One-Dimensional Trajectories

`quintic` (5th order) polynomial trajectory

`traj =quintic(q0,qF,M)` is a ***scalar trajectory*** (Mx1) that varies smoothly from q0 to qF in *M steps* using a ***quintic (5th order) polynomial***.

Position (`traj.q`), velocity (`traj.qd`), and acceleration (`traj.qdd`)

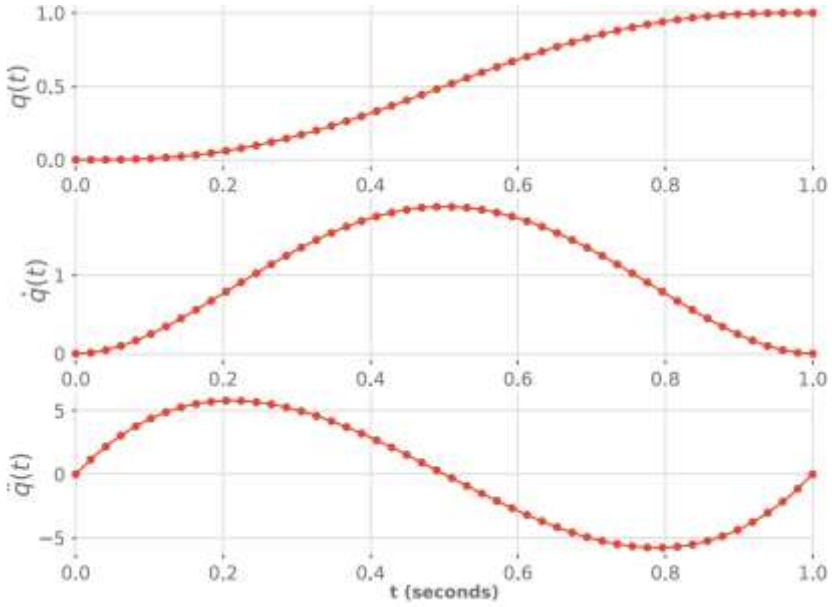
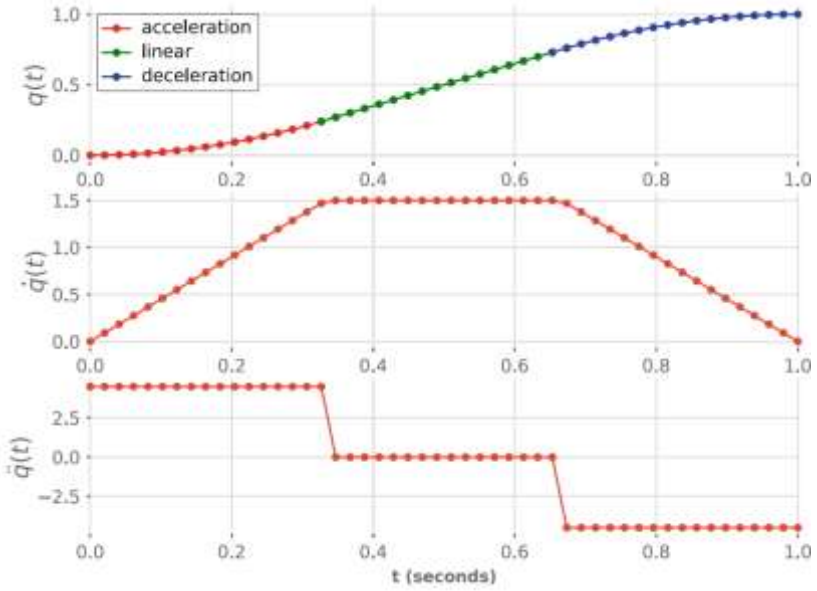


Fig. 3.2. Quintic polynomial trajectory. From top to bottom is position, velocity and acceleration versus time step. **a** With zero-velocity boundary conditions,

Smooth One-Dimensional Trajectories

`trapezoidal` Linear segment with parabolic blend

`traj=trapezoidal(q0,qF,M)` is a ***scalar trajectory*** (Mx1) that varies smoothly from S0 to SF in ***M steps*** using a ***constant velocity segment*** and parabolic blends (a trapezoidal path). It is continuous in position and velocity, but not in acceleration.



Multi-Dimensional Trajectories

- Most useful ***robots*** have more than ***one*** axis of motion and it is quite straightforward to extend the smooth ***scalar trajectory*** to the vector case.

- In terms of ***configuration space***, these axes of motion correspond to the dimensions of the robot's configuration space – to its Degrees of Freedom (DoFs).

DJI Phantom 4

Thorvald Robot

ABB YuMi Robot

Multi-Dimensional Trajectories

- We represent the ***robot's configuration*** as a vector $q \in \mathbb{R}^n$ where n is the number of degrees of freedom.
  a. Robot arm with 6-joints: $q = (q_1, q_2, q_3, q_4, q_5, q_6)$
  b. Wheeled mobile robot: $q = (x, y, \theta)$
  c. Aerial mobile robot: $q = (x, y, z, \theta_r, \theta_p, \theta_y)$
- In all these cases we would require ***smooth*** multi-dimensional motion from an initial configuration vector q_0 to a final configuration vector q_f.

Multi-Dimensional Trajectories

`mtraj` Multi-axis trajectory between two points

`[Q,QD,QDD]=mtraj(TFUNC,Q0,QF,M)` is a ***multi-axis trajectory*** (MxN) varying from configuration Q0 (1xN) to QF (1xN) according to the scalar trajectory function TFUNC in ***M steps***.

The **first argument** (`TFUNC`) is a function that generates a smooth *scalar* trajectory, e.g., `trapezoidal` or `quintic`

Multi-Segment Trajectories

- Need to move along a path trough one or more via-points without stopping:
  - Obstacle avoidance
  - Follow robot pose/targets through piecewise continuous trajectories
- Trajectory **q** defined by M via-points $\mathbf{q}_k$, $k \in [0, M-1]$, *i.e.*, M-1 motion segments
- Robot at rest at $\mathbf{q}_0$ and $\mathbf{q}_{M-1}$
- Moves through (or close to) other $\mathbf{q}_i$ without stopping

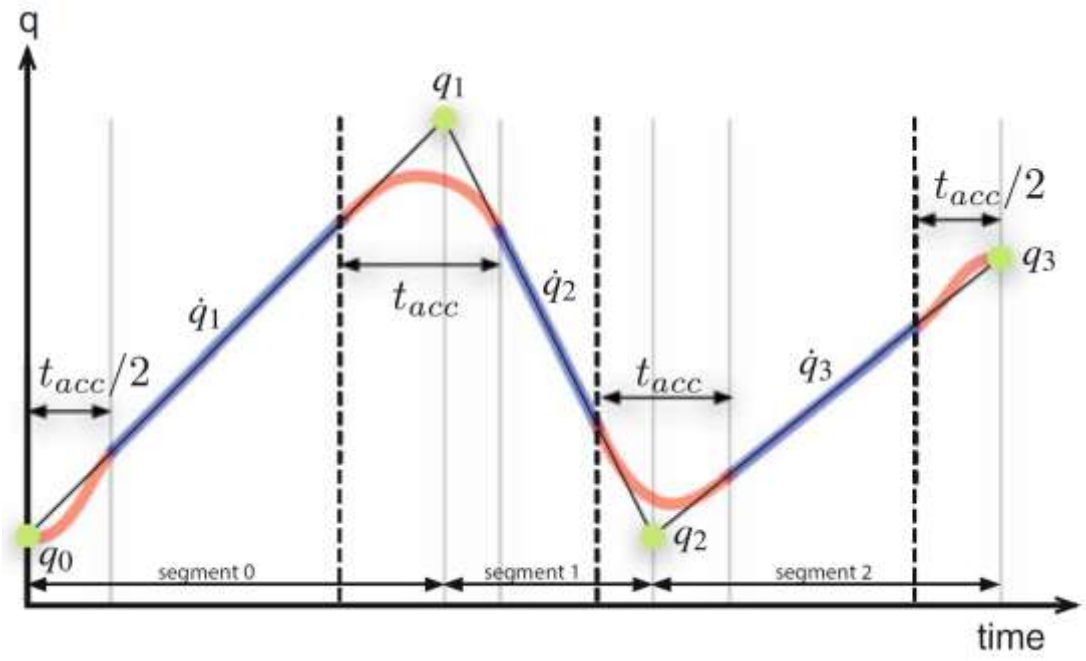


Fig. 3.5.
Notation for multi-segment trajectory showing four points and three motion segments. Blue indicates constant velocity motion, red indicates regions of acceleration

Interpolation of Orientation in 3D

- Important, e.g. if we require the end effector of a robot to smoothly change from orientation $\boldsymbol{\xi}_0$ to $\boldsymbol{\xi}_1$ in $\mathbf{S}^3$
- Want some function $\boldsymbol{\xi}(s) = \sigma(\boldsymbol{\xi}_0, \boldsymbol{\xi}_1, s)$ where $s \in [0, 1]$ which has the boundary conditions $\sigma(\boldsymbol{\xi}_0, \boldsymbol{\xi}_1, 0) = \boldsymbol{\xi}_0$ and $\sigma(\boldsymbol{\xi}_0, \boldsymbol{\xi}_1, 1) = \boldsymbol{\xi}_1$ and where $\sigma(\boldsymbol{\xi}_0, \boldsymbol{\xi}_1, s)$ varies smoothly for intermediate values of s.
- Linear interpolation of Rotation matrix $\boldsymbol{\xi} \sim \boldsymbol{R} \in \mathbf{SO}(3)$, should **not** be used as it typically result in not valid orthonormal matrix
- Instead, consider a representation $\boldsymbol{\xi} \sim \boldsymbol{\Gamma} \in \mathbb{S}^1 \times \mathbb{S}^1 \times \mathbb{S}^1$ and use

$$\sigma(\boldsymbol{\Gamma}_0, \boldsymbol{\Gamma}_1, s) = (1 - s)\boldsymbol{\Gamma}_0 + s\boldsymbol{\Gamma}_1$$

Wrapping up…

- There are ***many*** mathematical objects that can be used to represent pose and velocity of a rigid body:
  - Position: vector for 2D and 3D cases;
  - Orientation: angle, rotation matrix, and unit-quaternion for 2D and 3D cases;
  - Orientation and translation: homogeneous transformation matrix.
- 3D Matrices composition is ***non-commutative***, i.e., the order in which relative poses are multiplied is important:
  - World (or fixed) frame: pre-multiplication;
  - Body (or current) frame: post-multiplication.

Wrapping up…

- In ***robotics***, we often need to generate a ***time-varying pose*** that moves smoothly in translation and rotation;
- A ***path*** is a spatial construct that leads from an initial pose q_0 to a final pose q_f, whereas a ***trajectory*** is a path with specified timing and velocity;
- Here, we have studied 3 ***smooth trajectory generation*** methods for one dimension and multi-dimensional cases: `quintic`, `trapezoidal`, and `mtraj`;
- Indeed, `quintic` gives mean velocities far less than the ***maximum speed*** while `trapezoidal` gives a ***trapezoidal velocity*** profile with non-smooth accelerations.
- **Multi-segment trajectories** and **interpolation of orientations** are useful to know about for the upcoming robot-lab

Thank you for your attention !